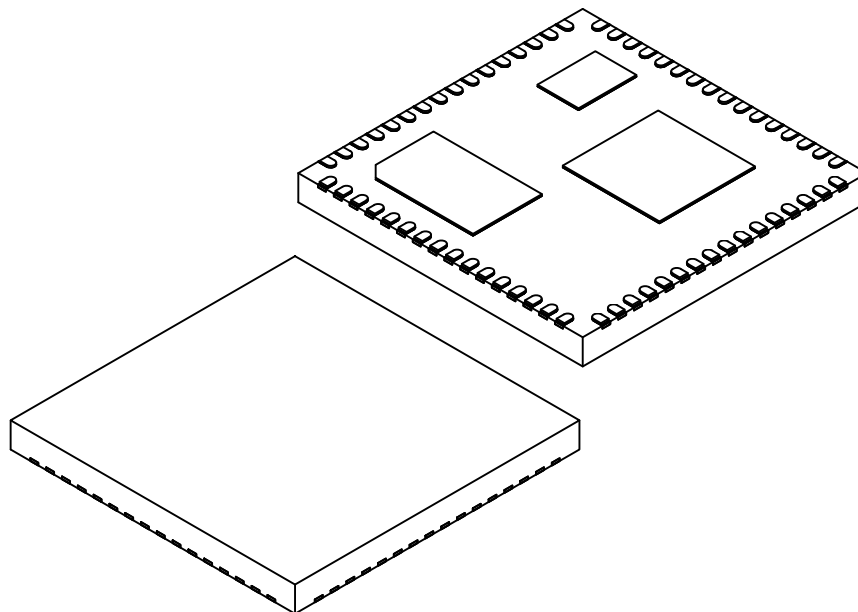


Packaging Diagrams and Parameters

64-Lead Very-Thin Grid-Array Quad Flatpack No-Lead Package (8QW) - 9x9x0.927 mm Body [VGQFN] with 3 Exposed Pads and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	0.827	0.877	0.927
Standoff	A1	0.00	-	0.05
Terminal Thickness	A3	0.127 REF		
Overall Length	D	9.00 BSC		
Overall Width	E	9.00 BSC		
Exposed Pad 1 Length	D1	2.08	2.18	2.28
Exposed Pad 1 Width	E1	3.33	3.43	3.53
Exposed Pad 2 Length	D2	2.92	3.02	3.12
Exposed Pad 2 Width	E2	2.96	3.06	3.16
Exposed Pad 3 Length	D3	1.73	1.83	1.93
Exposed Pad 3 Width	E3	1.20	1.30	1.40
Terminal Width	b	0.175	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Wettable Flank Step Cut Width	E4	0.01	0.05	0.09
Wettable Flank Step Cut Depth	A4	0.03	-	-

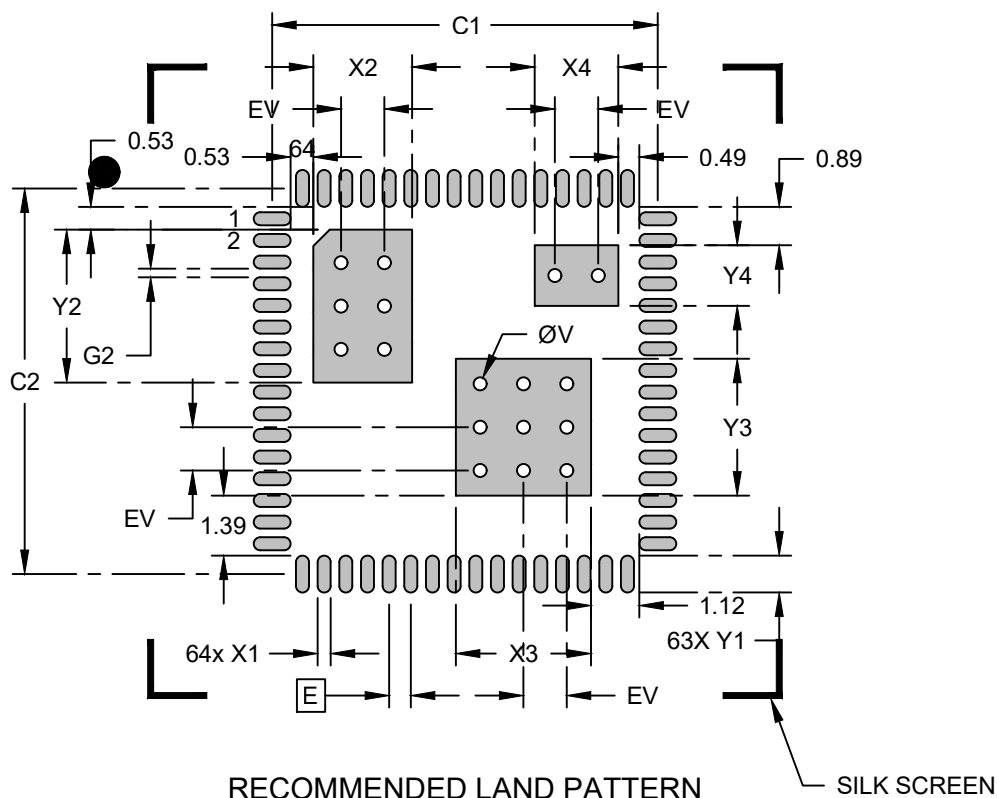
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
REF: Reference Dimension, usually without tolerance, is for information purposes only.

Land Pattern

64-Lead Very-Thin Grid-Array Quad Flatpack No-Lead Package (8QW) - 9x9x0.927 mm Body [VGQFN] with 3 Exposed Pads and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS			Units		MM
Dimension Limits		MIN	NOM	MAX	Dimension Limits		MAX
Contact Pitch	E	0.50 BSC			Center Pad 1 Width	X2	2.28
Contact Pad Spacing	C1		8.90		Center Pad 1 Length	Y2	3.53
Contact Pad Spacing	C2		8.90		Center Pad 2 Width	X3	3.12
Contact Pad Width (X64)	X1			0.30	Center Pad 2 Length	Y3	3.16
Contact Pad Length (X64)	Y1			0.85	Center Pad 3 Width	X4	1.93
Contact Pad to Contact Pad (X60)	G2	0.20			Center Pad 3 Length	Y4	1.40
Thermal Via Diameter	V		1.00				
Thermal Via Pitch	EV		0.30				

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, please refer to the current industry standard IPC-7093.